TANGI SARATH KUMAR

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.24
NRI JUNIOR KALASALA	Class XII (BIEAP)	2021	95.40%
GOVT HIGH SCHOOL	Class X (SSC)	2019	9.50%

SCHOLASTIC ACHIEVEMENTS

Secured 1st Rank at the Mandal Level in a Mathematics Competition during school

POSITION OF RESPONSIBILITY

Teaching Assistant: Teaching Assistant for Mechanics of materials which includes invigilation, evaluation and tutorial sessions for students at IITH and demonstration of experiments.

SKILLS

Software's: Autocad, Catia, Ansys workbench, Abaqus CAE, solidworks

Programming skills:python, MATLAB

Languages:English, Telugu , Hindi (moderate)

Soft skills: MS Office, communication skills

RELEVANT COURSES

Finite element method , Vibration and dynamics , Advanced mechanics of solids , Advanced topics in mathematical tools , vehicle dynamics , Machine learning and its applications

PROJECTS

Project 1: Nonlinear Dynamics of Serpentine Spring-Like Structures (AUG 2025-Present)

- **Engineered and simulated** serpentine spring structures using **FEA**.
- **Characterized nonlinear force-displacement and dynamic response** under large deformation.
- Validated Abaqus simulation results against experimental data to evaluate the nonlinear force-displacement behavior of the serpentine spring.

Tools & Skills: Abaqus/CAE , FEA , MATLAB

Project 2: Nonlinear Post buckling Analysis of a Riveted Stiffened Aircraft Fuselage Panel using COMSOL Multiphysics(MAY 2026 - JULY 2026)

- Developed a **nonlinear shell FE model** of a **pressurized Al 6063-T83 fuselage panel** with **skin, frames, stringers and 6 mm riveted joints**, using **contact and fastener modelling**.
- Simulated **combined pressure and axial compression** (load-ratio sweep) to capture **skin buckling and post buckling response**, with **onset at k 0.6**.
- Assessed **rivet integrity** with a **combined normal-shear failure criterion (safety factor 4)**, locating **peak rivet loads (~870 N shear)** at the **frame-stringer junction**.

Tools & Skills: COMSOL Multiphysics , FEA

Project 3: Hydrodynamic Performance & Cavitation Analysis of Marine Propellers (Dec2024-April2025)

- Focused on **comparative analysis** of **3-blade**, **4-blade**, and **5-blade** ship propellers.

- **Evaluated thrust generation, pressure distribution, and cavitation risk** across different blade configurations.
- **Validated the CFD setup** against **Watanabe et al. experimental data**, with the **K_T-J slope within 1.5%**, and **recommended the 4-blade design** as the best **thrust-to-cavitation trade-off**.
- Compared **CFD** results to assess the effect of blade count on marine propeller performance.

Tools & Skills: SolidWorks **,** ANSYS CFD , MATLAB

EXPERIENCE

ONGC, Tatipaka : (Intern) (JUNE 14 2023- JUNE 28 2023)

- Gained basic exposure to **natural gas processing and compression operations** at the ONGC Tatipaka facility.
- Observed the working of **gas compressors, separators, intercoolers, and other process equipment**.
- Learned the fundamentals of **gas compression, pressure control, temperature monitoring, and plant operations**.

Visakhapatnam Steel Plant (RINL) : (Intern) (MAY 06 2024- JUNE 01 2024)

- Gained basic exposure to **integrated steel manufacturing processes and plant operations**.
- Observed major industrial facilities including **Blast Furnace, Coke Oven, Sintering Plant, Foundry, and Thermal Power Plant**.
- Learned the basic functioning of **steel-making equipment, material processing, casting, and thermal systems**.
- Studied the **material flow from raw materials to finished steel** across the plant's major **production units**.

EXTRACURRICULAR ACTIVITIES

Playing cricket , Watching movies , and learning new mechanical softwares